

Yunshu Chen

AI & Software Engineer · Clinical AI, Product Development, Healthcare Data

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Machine learning engineer and PhD candidate with four years building and delivering software and ML systems, from a mobile app to medical imaging tools. My work with hospital and industry partners includes cell-analysis software in use at Lead Healthcare, an IVUS reporting agent undergoing cardiologist review, and a clinical data pipeline for Western Health.

EXPERIENCE

Machine Learning Engineer (Part-time)

Jan 2026 – present

PredictX Health × Western Health, Melbourne. Chemotherapy toxicity prediction for colorectal cancer

- Built a clinical data pipeline combining registry, CT-derived body composition and chemotherapy records for a **302-patient cohort**. Added pytest regression tests and a data dictionary for clinical collaborators; reviewed progress and data questions in **weekly meetings with Western Health**.
- Identified and traced **chemotherapy-regimen errors affecting eight patients** and a dosing entry error before modelling. Traced the regimen errors to a script reading an intermediate file instead of the master table and documented corrections for audit.

Machine Learning Researcher / Engineer (PhD)

Oct 2023 – present

Monash University, with the Victorian Heart Institute, Lead Healthcare and Apollo Medical Imaging

- Cell-analysis software, Lead Healthcare.** Developed a segmentation model trained on **100K+ images and 8M+ cell instances**, reaching 89% mAP. Packaged and delivered it as software now in use at Lead Healthcare; built a **React interface and an API for internal laboratory use**. LoRA fine-tuning supports new cell types from 5–10 samples.
- IVUS reporting agent.** Built a hybrid workflow in which an LLM coordinates measurement tools and drafts reports from their outputs. Used **JSON Schema tool definitions, Pydantic state records and retrieval of prior clinician feedback**; eight decision tools include guideline citations. Runs on **local Qwen2.5-7B with vLLM**. Built the review interface and prepared **35 cases**; **cardiologist review is ongoing**.
- IVUS measurement and data engineering.** Designed evaluation around agreement with expert measurements from the outset (**CCC 0.94 / 0.96** for lumen / vessel area). Distilled the model to **2.6× smaller and 3× faster on CPU** without reducing measured accuracy. Built a separate **49.6M-frame DICOM ingestion pipeline** with versioned data lineage for ultrasound pre-training.
- Microscopy tools and feasibility.** Delivered a **PyQt zebrafish analysis interface** to biologists. For cell tracking, evaluated five alternative approaches before using SAM2 memory to link cells without retraining (**CHOTA 0.860 vs 0.851**); documented unsuccessful approaches and their limitations.

Speech Recognition Algorithm Engineer, Intern (Full-time), NIO Intelligent Cockpit, Shanghai

Jul 2022 – Dec 2022

- Improved multilingual ASR accuracy by **5%** across Chinese and European languages; built automated diagnostics that resolved **60+ production recognition failures**; **integrated models into production vehicle systems** alongside product, UX and NLU teams.

SELECTED PROJECTS

- Genius Labbook** (open source, github.com/CHEN-Yunshu/Genius_Labbook). Python CLI logging experiments, results, figures and decisions as Markdown + YAML with automatic git commits, with **documented Claude Code integration** (a hook and a slash command); **135 entries across 6 concurrent projects** since March 2026.
- End-to-Same-End Encryption app** (honours thesis, 2023). Cross-platform iOS/Android encrypted storage where the server never sees plaintext; owned the full stack: **React Native** front end, **Spring Boot and Node.js** backend, **AWS S3, Jenkins CI/CD**.

TEACHING & LEADERSHIP

- Sessional Teaching Assistant, Monash University** (2025 – present): weekly labs for **60+ postgraduates**, 92/100 satisfaction. **General Executive, SRC, and student member of the Academic Board**, University of Sydney (2021 – 2022).

EDUCATION, PUBLICATIONS & AWARDS

PhD, Data Science & AI, Monash University. Thesis in writing stage, available full time

Oct 2023 – Apr 2027 (expected)

Bachelor of Advanced Computing, Honours II-1, The University of Sydney. Major: Software Development

Feb 2019 – Jun 2023

Three first-author papers under review in medical imaging and computer vision; co-author of an IEEE ISCSLP 2022 paper. Our team placed **2nd** and **3rd** in the two tracks of the Intelligent Cockpit Speech Recognition Challenge, Singapore 2022.

TECHNICAL SKILLS

AI orchestration	ReAct agents · JSON Schema · RAG · Pydantic · vLLM · Qwen (local) · DeepSeek / Gemini (APIs)
Machine learning	PyTorch · scikit-learn · SAM / SAM2 · nnU-Net · LoRA · knowledge distillation · Random Forest / LASSO
Software & APIs	Python (advanced) · Java · SQL · React / React Native · Spring Boot · Node.js · pytest · Python packaging
Cloud & DevOps	AWS S3 · Docker · Jenkins CI/CD · Git · Linux · versioned data lineage
Healthcare data	DICOM · clinical registries · CT, MRI, IVUS · validation with CCC, Bland–Altman, Cohen's κ, AUROC
AI-assisted dev	Claude Code daily (hooks, slash commands) · GitHub Copilot · Cursor